

STAT 24410/30030, Homework 2
Due Thursday, October 17, at 2:00pm

1. (10 pts) Lecture Notes, Chapter 2, Exercise 2.
2. (10 pts) Lecture Notes, Chapter 2, Exercise 3.
3. (10 pts) Lecture Notes, Chapter 2, Exercise 4.
4. (15 pts) Lecture Notes, Chapter 2, Exercise 6.
5. (15 pts) Lecture Notes, Chapter 2, Exercise 7.
6. (15 pts) Lecture Notes, Chapter 2, Exercise 8.
7. (15 pts) Let X_1 and X_2 be independent random variables $\sim \text{Exp}(\lambda)$.
 - (i) Find the cumulative distribution function and density function of $R = \max(X_1, X_2) - \min(X_1, X_2)$.
 - (ii) Prove that $\min(X_1, X_2)$ and $X_1 - X_2$ are independent random variables.
8. (10 pts) The goal of this exercise is to use simulations to understand dependence between random variables. We are in the setting of Problem 7.
 - (i) Generate 500 samples of 2 random variables X_1 and X_2 with distribution $\text{Exp}(1)$. For each of the 500 samples, calculate $\min(X_1, X_2)$, $\max(X_1, X_2)$, $R = \max(X_1, X_2) - \min(X_1, X_2)$, and $(X_1 - X_2)$. This should give you 500 tuples/arrays, each of them constructed using 2 independent draws. Create a scatter plot of $X_1 - X_2$ versus $\min(X_1, X_2)$ and comment on if it corresponds to what you proved in Problem 8.
 - (ii) Create two scatterplots to investigate (visually) if: (a) X_1 is independent of $\min(X_1, X_2)$; (b) $\max(X_1, X_2)$ is independent of $X_1 - X_2$. Comment on how the scatterplots can inform you on ideas for a proof.